

Python Code

```
import requests
import pandas as pd
import re
from bs4 import BeautifulSoup
from affinn import Affin
from nrclex import NRCLex

# Q1 - On "https://www.commonsensemedia.org/book-reviews" search for "kids-books-about-911".

url = "https://www.commonsensemedia.org/search/category/book/sort/score-desc/kids-books-about-911"

# Q2 - 2- Get the URL for the first page of results and utilize Beautiful Soup to get all of reviews from these pages.

def beautiful_soupify(url):
    """Fetches the HTML content of a given URL and parses it using BeautifulSoup.

    Args:
        url (str): The URL of the web page to fetch.

    Returns:
        BeautifulSoup object or None: The parsed BeautifulSoup object if the request is successful, or None if the request fails.
    """

    # Send a GET request to the specified URL
    response = requests.get(url)

    # Check if the request was successful (status code 200)
    if response.status_code == 200:
        # Parse the HTML content using BeautifulSoup
        soup = BeautifulSoup(response.content, 'html.parser')
        return soup
    else:
        print("Connection request failed. Status code:", response.status_code)
        return None # Return None to indicate a failed request
```

```

def extract_books(url):
    """Extracts books from a given Common Sense Media search URL.

    Args:
        url (str): The URL of the page to scrape.

    Returns:
        ResultSet: A BeautifulSoup ResultSet with URLs for each book found in the search.
    """

    soup = BeautifulSoup(url)

    if soup:
        # Find all links with href containing "/book-reviews/" and no class attribute
        # This retrieves a link to each book on the webpage only once
        links = soup.find_all('a', attrs={'href': re.compile("/book-reviews/"), 'class': ""})

    return links

def extract_reviews(url):
    """Extracts reviews from a given Common Sense Media book review URL.

    Args:
        url (str): The URL of the page to scrape.

    Returns:
        list: A data frame with book name and reviews.
    """

    reviews = []

    # Extract parent reviews
    adult_url = "https://www.common Sense Media.org" + url + "/user-reviews/adult"
    adult_soup = BeautifulSoup(adult_url)
    title = adult_soup.find('h1').text
    result = adult_soup.find_all("div", class_="reveal__content")

```

```

for res in result:
    reviews.append({"book title": title, "review": res.text})

# Extract kid reviews
child_url = "https://www.commonensemedia.org" + url + "/user-reviews/child"
child_soup = BeautifulSoup(child_url)
result = child_soup.find_all("div", class_="reveal__content")
for res in result:
    reviews.append({"book title": title, "review": res.text})

return pd.DataFrame(reviews)

def get_all_reviews(url):
    """Gets reviews for all books from a given Common Sense Media search URL.

    Args:
        url (str): The URL of the search page.

    Returns:
        pd.DataFrame: A DataFrame containing book titles and reviews.
    """

    # Initialize an empty DataFrame to store reviews
    reviews = pd.DataFrame({"book title": [], "review": []})

    # Extract book links from the search URL
    for link in extract_books(url):

        # Extract reviews for the current book
        book_reviews = extract_reviews(link["href"])

        # Concatenate the book's reviews with the existing DataFrame
        reviews = pd.concat([reviews, book_reviews])

    return reviews

```

```
# # Q3 - Use afinn method for sentiment analysis.  
# The output should be a data frame with each review, the afn.score and label you create based on the score (0:  
neutral, <0 negative and >0 positive).
```

```
def afinn_sentiment_analysis(review):  
    """Performs sentiment analysis on a given review using the AFINN lexicon.
```

Args:

review (dict): A dictionary containing the 'review' text.

Returns:

tuple: A tuple containing the sentiment score and label.

```
    """
```

```
# Create an AFINN analyzer
```

```
afinn = Afinn()
```

```
# Extract the review text
```

```
text = review['review']
```

```
# Calculate the sentiment score
```

```
sentiment_score = afinn.score(text)
```

```
# Assign a sentiment label based on the score
```

```
if sentiment_score < 0:
```

```
    label = 'negative'
```

```
elif sentiment_score == 0:
```

```
    label = 'neutral'
```

```
else:
```

```
    label = 'positive'
```

```
return sentiment_score, label
```

```
# Get reviews for all books from the search URL
```

```
afinn_reviews = get_all_reviews(url)
```

```
# Perform sentiment analysis using AFINN for each review
```

```
afinn_reviews[['afinn_score', 'label']] = afinn_reviews.apply(
    afinn_sentiment_analysis, axis=1, result_type="expand"
)
```

```
# Print the DataFrame with sentiment scores and labels
print(afinn_reviews)
```

```
# # Q4 - Add the name of the restaurant to the data frame in question 3.
```

```
"""
```

It was much easier to add the name of the book to the dataframe when extracting the reviews, so this has already been done in the extract_reviews function.

```
"""
```

```
# # Q5 - Use NRCLEX method for sentiment analysis.
```

```
# Pass each review to NRCLEX and get top_emotions for that review, store all reviews and their top emotions in a text file.
```

```
def NRCLEX_sentiment_analysis(review):
```

```
    """Performs sentiment analysis on a given review using the NRC Lexicon.
```

```
    Args:
```

```
        review (dict): A dictionary containing the 'review' text.
```

```
    Returns:
```

```
        dict: A dictionary containing the top emotions and their associated scores.
```

```
    """
```

```
    text = review['review']
```

```
    text_object = NRCLEX(text)
```

```
    return text_object.top_emotions
```

```
# Get reviews for all books from the search URL
```

```
NRCLEX_reviews = get_all_reviews(url)
```

```
# Perform sentiment analysis using NRC Lexicon for each review
```

```
NRCLEX_reviews['emotions'] = NRCLEX_reviews.apply(
```

```
    NRCLEX_sentiment_analysis, axis=1
```

```
)

# Drop the 'book title' column as it's not needed for output
NRCLex_reviews = NRCLex_reviews.drop('book title', axis=1)

# Specify path for export
path = r'/Users/jirissman/Documents/coursework/BU/CS688/Assignment 3/NRCLex_emotions.txt'

# Export DataFrame to text file
with open(path, 'a') as f:
    f.write(NRCLex_reviews.to_string(index=False, header=False))
```

Console Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● jirissman@Josephs-MacBook-Air ~ % /usr/local/bin/python3 "/Users/jirissman/Documents/coursework/BU/CS688/Assignment 3/Rissman_Hw3.py"
    book title          review  afinn_score  label
0  Nothing Like the Movies: Better Than the Movie... \n  AMAZING BOOK! Many pop culture reference...      8.0  positive
1  Nothing Like the Movies: Better Than the Movie... \n  i actually loved this book so much (low ...    21.0  positive
0          Diary of a Wimpy Kid: Hot Mess \n  just.... PERFECT I cant even discribe i...      8.0  positive
1          Diary of a Wimpy Kid: Hot Mess \n  The 19th book in the famous Wimpy Kid se...      3.0  positive
2          Diary of a Wimpy Kid: Hot Mess \n  AMAZING BOOK!!!! I ABSOLUTELY LOVE THE ...      4.0  positive
3          Diary of a Wimpy Kid: Hot Mess \n  \n This book is just too good.\n      3.0  positive
0  Wrath of the Triple Goddess: Percy Jackson and... \n  I liked this book, and I think any kids ...      0.0  neutral
0          The Book of Bill \n  I got this book for my birthday, and it'...    18.0  positive
0  The Graveyard Gift: Fern's School for Wayward ... \n  As a father of three in the U.S., I've a...    13.0  positive
○ jirissman@Josephs-MacBook-Air ~ %
```

NRCLex Text File Output

\n AMAZING BOOK! Many pop culture references from shows like "Gilmore Girls" one of my personal favorites. Some inappropriate moments but teenagers should be fine.\n [(negative, 0.25)]

\n i actually loved this book so much (low key better than the first but don't come for me). i loved how wes never gave up on liz and kinda loved how grown up liz had gotten! and oh my gosh i literally love sarah and clark together i need a book of just them two!! lynn painters books are always so good to me because as a young christian it's always hard to find a clean romance book, but lynn's have never let me down. i will say i sobbed my eyes out the whole time i was reading this because of what wes had to go through and losing liz, but if you cry while reading a book doesn't that just show how good it is?! anyways 10/10 would recommend! :) \n [(positive, 0.2564102564102564)]

\n just.... PERFECT I cant even discribe it its too good to be true\n
[(trust, 0.25), (positive, 0.25), (joy, 0.25)]

\n The 19th book in the famous Wimpy Kid series is great. It features a ton of the Heffley family drama. Expect some conversations about dating and a conversation on Susan and her sister fighting over a boyfriend.\n
[(anger, 0.25), (trust, 0.25), (positive, 0.25), (negative, 0.25)]

\n AMAZING BOOK!!!! I ABSOLUTELY LOVE THE DIARY OF A WIMPY KID SERIES!!!! I think this is for a little bit older kids, because family issues shouldn't have younger kids worried!\n
[(sadness, 0.5)]

\n This book is just too good.\n
[(trust, 0.2), (surprise, 0.2), (positive, 0.2), (joy, 0.2), (anticipation, 0.2)]

\n I liked this book, and I think any kids 9+ who enjoy Percy Jackson, or fantasy should read it, it doesn't have any swearing, and the only elements of violence are the main characters fighting monsters, so I belive this is a great book for kids aged 9+.\n
[(anger, 0.181818181818182), (positive, 0.181818181818182), (negative, 0.181818181818182)]

\n I got this book for my birthday, and it's my favorite thing. If you or your child are fans of Gravity Falls, I think you'll agree with me. There are so many lore drops, little details that make the book so much more interesting, pop culture and history references, and the same quirky, kinda gory out-of-pocket humor that we all love so much because of the show. \n\nAlbeit there is gore and "bad life advice" (remove your skin, mix together all the expired fluids in your fridge and drink them, ect...), it's all pretty obviously satire. However, I won't buy this book for an elementary schooler or younger, because young, impressionable minds might not be able to tell the difference between satire and non-active.\n\nBut if your kid has watched Gravity Falls and is a fan, I think that they will LOVE this book!\n
[(positive, 0.20408163265306123)]

\n As a father of three in the U.S., I've always been the type to plan. My wife and I had built a solid financial future for our kids, and a huge part of that was tied up in Bitcoin. I had \$750,000 saved in a hardware wallet, and I thought it was the safest place to store it. I

made sure everything was secure because this wasn't just money—it was my kids' future. I didn't take risks when it came to that. But then the unthinkable happened. One day, after a routine software update, my wallet froze. It wouldn't let me access anything. My heart dropped. This was our nest egg, and I was suddenly locked out of it. I tried everything: reboots, troubleshooting, and reaching out to the wallet's customer support but nothing worked. I felt like I had hit a brick wall, and the panic set in. I couldn't afford to lose that money. It wasn't just mine; it was my family's. Desperate, I started researching recovery services. I wasn't sure if I could trust any of them. There are so many scams out there, and I couldn't afford to make another mistake. That's when I found Trust Geeks Hack Expert. Their reviews were overwhelmingly positive, and it seemed like they had dealt with situations exactly like mine before. I decided to give them a shot. From the first contact, they were professional and calm, which immediately eased my mind. They asked me all the right questions and wasted no time getting to work. The process took just a few days, but it felt like a lifetime. I kept holding my breath, waiting to hear if they had succeeded. When I finally got the message that they had recovered my funds, I didn't know whether to laugh or cry. I logged into my wallet, and there it was: my entire Bitcoin balance, just as it had been. The relief I felt was indescribable. It was like a huge weight had been lifted off my shoulders. don't hesitate— Contact Trust Geeks Hack Expert At

<https://trustgeekshackexpert.com/> --Email: Trustgeekshackexpert{@}fastservice{.}com,\n [(positive, 0.2375)]

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\n i actually loved this book so much (low key better than the first but don't come for me). i loved how wes never gave up on liz and kinda loved how grown up liz had gotten! and oh my gosh i literally love sarah and clark together i need a book of just them two!! lynn painters books are always so good to me because as a young christian it's always hard to find a clean romance book, but lynn's have never let me down. i will say i sobbed my eyes out the whole time i was reading this because of what wes had to go through and losing liz, but if you cry while reading a book doesn't that just show how good it is?! anyways 10/10 would recommend! :) \n [(positive, 0.2564102564102564)]

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